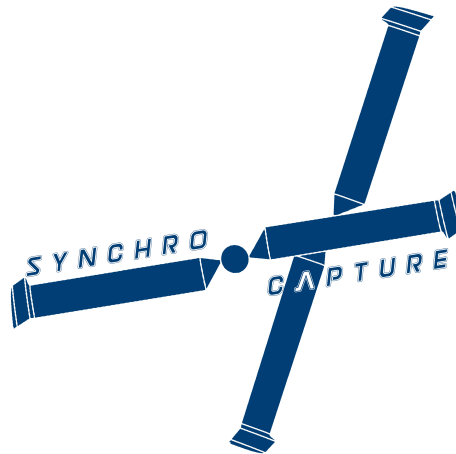


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Fuselage Structural Design



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Abstract

A novel helicopter fuselage truss design, inspired by results from Generative Design in Fusion 360 is explored. Using a simple welded tubular construction manufactured out of Al 7075 results in an easy and cheap to manufacture airframe that is lightweight, yet meets all FAR 29 requirements.

A tubular landing gear is designed using an equivalent static load factor, and is optimized to yield under heavy landings to maximise energy absorption, and minimise shock loading to the fuselage. In the end, a structure weighing 130 kg is obtained, significantly lower than the mass predicted during the initial sizing phase.

Acknowledgements

To my lovely WRR03 team members, a hearty thanks, for without you I would not have been able to complete this project, so many skills acquired, and so many memories created. I would like to also extend my gratitude to the project coordinators, who have done an absolutely splendid job managing us. Lastly, huge thanks to our supervisor Dr. Zahra, for always helping us out and providing useful feedback.

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Table 1: *Team Members and Roles.*

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1 Introduction

The aim of the *SynchroCapture* Project is to design an aerial recovery system for a rocket capable of inserting a payload into LEO. Developing such a system that recovers 100% of a rocket could lead to a significant reduction in the cost of carrying payloads to orbit as the recovered stages can be reused with minimal refurbishment effort. The complete design requirements are listed in the brief [2].

1.1 Introduction to WRR03

The entire project was divided into 6 sub teams, each focusing on a particular area of conceptual and preliminary design of the aircraft. The responsibilities of WRR03 were:

1. Create an internal layout of all major sub systems of the aircraft.
2. Structural Design and Material selection of the aircraft, ensuring regulations and mission constraints are met.
3. Selection of a manufacturing methodology for the airframe.
4. Creating a CAD model of the aircraft to visualise the proposed concept.

2 Concept Selection in Week 1

As the project brief did not specify a recovery method or aircraft concept, the group had considerable freedom to select a platform and baseline configuration. Three main concepts were considered, and the entire group was split into 3 smaller groups, to explore a fixed wing, helicopter, and a VTOL configuration respectively. The author of this report was assigned to the Fixed-wing sub group, where he was responsible for conceptualizing a recovery method and obtaining an initial cost estimate for pursuing a fixed-wing recovery vehicle.

2.1 Fixed-Wing recovery method concept

One of the concepts proposed by the Fixed-Wing sub group was a mothership concept similar to the White Knight aircraft [3], however instead of releasing a spaceplane, it would recover a gliding rocket stage. This concept is illustrated in Fig. 1

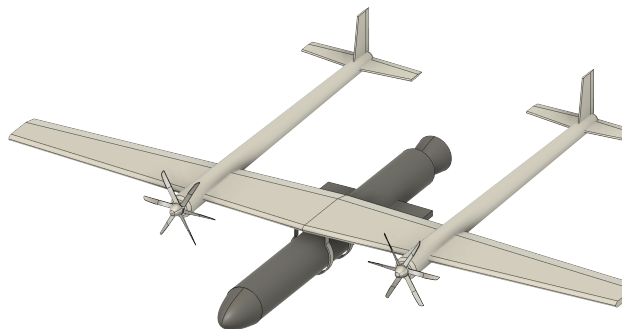


Figure 1: CAD model of a mothership recovery concept. Note that the gliding rocket stage is highlighted in dark grey.

This concept was unfortunately rejected, as it required the rocket stage to have a decent glide slope, as well as exceptional control authority in the gliding state. Furthermore, a precise matching of trajectory between the gliding stage and the recovery aircraft would be required. Not only would this make the control design of both the aircraft and rocket stage extremely challenging, but would also pose a significant safety hazard as the two aircraft are so close together.

2.2 Cost Estimation

For a complete comparison between the Fixed-Wing, Helicopter, and VTOL concepts, cost estimates for one prototype, and the variation of cost with number of aircraft manufactured was required. To calculate the costs for a fixed-wing aircraft, the method highlighted in [4] was used. This method consists of several empirical relations based on a large dataset of commercial and military aircraft. The cost is highly dependent on key aircraft parameters and labour costs, summarised in Table 2. Labour costs are 1989 estimates provided by [4], multiplied by the cost escalation factor (CEF) between 2020 and 1989. A more detailed discussion regarding initial aircraft parameters can be found in [5].

Labour	1989 Rate [\$ /hr]	2020 Rate [\$ /hr]	Aircraft Parameter	Value
Engineering	60	123	Take-off Weight	8500 kg
Manufacturing	35	73	Cruising Speed	121 m/s
Tooling	45	92	Wing Area	62.5 m ²
(a) Labour Costs			(b) Aircraft Parameters	

Table 2: Key Cost Estimation Parameters

In this calculation, the cost of avionics and engines was assumed to be \$10 Million. The cost of avionics was over-estimated as the capture mechanism and precise control required for the capture manoeuvre might require expensive hardware. Furthermore, the effects of interest rates and the costs of financing an aircraft development program were ignored. The total cost of the development program was calculated to be approximately **\$148 Million**.

3 Baseline Configuration

Ultimately, the helicopter concept was selected as it had the lowest risks, and the highest technology readiness level compared to the fixed wing and VTOL concepts. To maximise the lifting capacity of the helicopter, a synchro-copter configuration with inter-meshing blades was used. This removed the need for a tail rotor, resulting in a short and compact tail. Furthermore, given the inherent risk of recovering a falling rocket, it was decided that the helicopter would be unmanned and controlled by a pilot on the ground. These decisions resulted in a very unconventional design, with a significant amount of design freedom. Fig. 2 shows the baseline configuration of the helicopter. Baseline parameters are summarised in Table 3.



Figure 2: Initial Render of the design.

Baseline Parameter	Mass [kg]
Design MTOW	2907
Empty Weight	1229
Fuel Weight	278
Payload Mass	1400

Table 3: Baseline parameters

4 Fuselage Internal Layout

Before commencing the structural design of the helicopter, the internal layout of the fuselage had to be determined. Given the unmanned nature of the helicopter, it did not require a large cockpit in the front. Thus, the only requirement for the internal layout was that the c.g. of the entire helicopter should lie at approximately the same longitudinal position as the rotors.

Initial estimates of component weights were calculated by the Systems sub-team, and are listed in TABLE. The methodology for weights estimation is explained in greater detail in [6] and [7]. Once the weights were obtained, their moment arms were evaluated using a spreadsheet to calculate the overall c.g from the rotor datum. During this layout phase, as the fuselage structure was not designed yet, it was assumed that the c.g. of the fuselage and landing gear structure would be very close to the position of the rotors.

To minimise fuselage drag and structural weight, the layout was carefully selected to be as compact as possible while keeping the longitudinal c.g. at the rotors. There were several other considerations with regards to component placement:

- To avoid c.g. shifts upon depletion of fuel, the fuel tank c.g. had to be placed at the helicopter c.g.
- The tail lever arm was fixed at 4 m by the aerodynamics team. [8]
- To prevent instability during payload capture and cruise with a slung payload, the payload attachment point had to be placed at exactly the same longitudinal position as the rotors, to prevent large pitching moments. Furthermore, to enhance stability in cruise, the Flight Mechanics team desired to have the vertical position of the payload attachment point as close as possible to the vertical c.g. of the helicopter without payload.

The layout that satisfied all these constraints and was finally selected is illustrated in Fig. 3. Note that Fig. 3 is using estimates of component dimensions, which were subsequently modified as the design progressed in the later weeks.

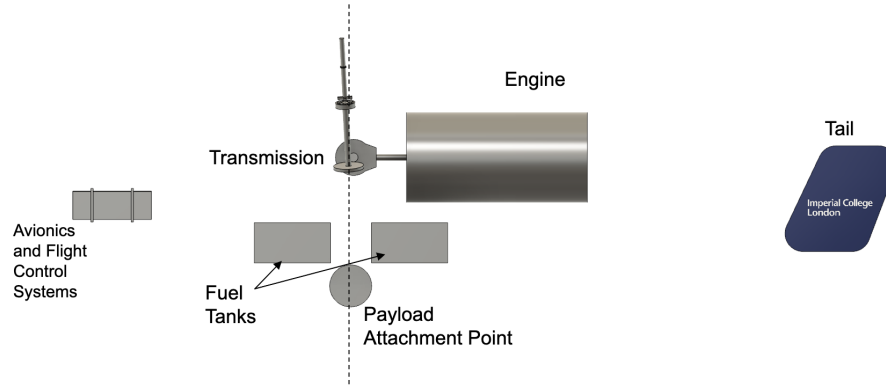


Figure 3: Fuselage Internal Layout

5 Fuselage Structural Design

5.1 Fuselage Loads

FAR 29 regulations stipulated a number of load cases to be satisfied with a safety factor of 1.5: [9]

1. A load factor of 3.5 to -1 g without payload. As the component weights were known this load case was easy to evaluate.
2. A load factor of 1.5 g with maximum payload. The load produced by the capture and cruise with an underslung payload were modelled by the systems team. [10]
3. Landing loads caused by a 10 ft/s descent rate. The load factor was estimated to be 3.5, using a method explained in Section 5.4.3
4. Asymmetrical tail loading with 100% of maximum loading in a symmetrical flight conditions acting on the surface on one side of the axis of symmetry. Tailplane loads were calculated by the Aerodynamics team. [8]

5.2 Structural Configuration

Conventional manned helicopters traditionally have had semi-monocoque airframes, which consist of frames and bulkheads to mount components, and a stressed skin construction with stringers to carry torque and bending loads. [11] Such a structural layout can be very efficient if a large internal volume is required, such as a cockpit, passenger compartment, or internal cargo space. However, given the lack of both cockpit and internal cargo volume requirements, it was decided that the fuselage would have a primary tubular truss structure, with a lightweight non load bearing skin structure to provide an aerodynamic shape. Such a configuration afforded several benefits over a semi-monocoque construction:

- Low cost manufacturing: As only a few truss members would be required for the airframe, construction of a truss structure would be considerably cheaper and faster compared to a semi-monocoque construction which would require substantial effort in machining frames, manufacturing stringers, and ultimately riveting all the airframe components. In comparison, a truss structure would be comprised of several tubes simply welded together.

- Improved accessibility: As the outer skin is non load bearing, it can be easily removed for maintenance, and truss members would not hinder access to internal components. This would make both construction and maintenance cheaper and easier.
- Improved aerodynamic performance: As the outer skin is non load bearing, it can be shaped exclusively with minimal trade-offs to minimise drag and/or to achieve certain characteristics (positive lift, pitching moment characteristics, etc.)

5.3 Truss Design

Once the configuration was decided, the truss geometry and member placement had to be designed. Given the multiple load cases, as well as the 3D structure of the truss, analytical methods proved to be extremely challenging given the time frame of the project. Thus, to obtain an efficient truss structure, the optimization tool Generative Design in Fusion 360 was used to get a visual representation of the load paths in the structure, as well as an initial reference geometry.

5.3.1 Component Mounting

Light frames were used to mount each sub-component to the truss structure, a method similar to the semi-monocoque design. The component would then bolt into the frame using tabs to keep it secure. In order to ensure stability, each component was held using two frames, placed fore and aft of the component centre respectively. These frames served not only as component mounts, but also as truss members in some regions. The geometry of these frames was not optimized yet, and they were simply modelled as having a constant rectangular cross section for initial truss design. These frames are highlighted in green in Fig. 5.

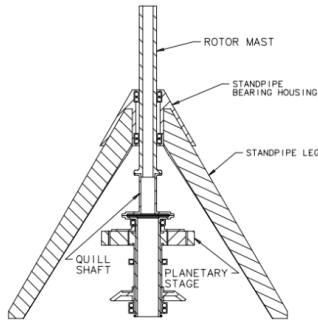


Figure 4: Standpipe-leg structure

A standpipe-leg structure illustrated in Fig. 4 was used for transferring the lift loads from the rotor to the fuselage truss. The thrust bearings in this configuration isolated the torque required to spin the rotors from the rotor lift, which ensured that the gearbox did not experience any vertical lift loading, simplifying its design.

Lastly, the payload attachment point was modelled as a ring onto which the payload slung line could be hooked onto. Given that the slung line and capture system would contain the necessary damping mechanisms to minimise shock loads upon aerial capture, the payload attachment point on the fuselage could be a very simple structure from a structural perspective.

5.3.2 Generative Design Study

Generative Design in Fusion 360 is a study that makes use of genetic and topological optimization algorithms, as well as artificial intelligence to design a structure that fulfills a set of requirements specified by the user while keeping mass at a minimum. Thus, it was used to obtain an insight into the primary load paths of the structure and to obtain a starting point for truss geometry.

To run the study, preserve and obstacle regions had to be modelled first. Preserve regions are the set of points where the study does not attempt to remove any material. The preserve regions in this case were the frames that the components were mounted to. Obstacle regions are the set of points that the study cannot enter, as they contain other components. In this case, the obstacle regions were preliminary

models of the components themselves. These regions are illustrated in Fig. 5, where the red features are obstacle regions and the green areas are preserve regions. The study essentially aimed to connect the preserve regions with as little material as possible.

Loads were applied to the preserve regions, by applying a constant pressure distribution on the respective component frames, and by adding a fixture onto the rotor thrust bearings for the flight cases. The landing load case was not entered into this study, as the landing gear was not modelled yet. Instead, it was decided that once the truss structure was designed, it would simply be tested with a landing load case to ensure it had satisfactory results.

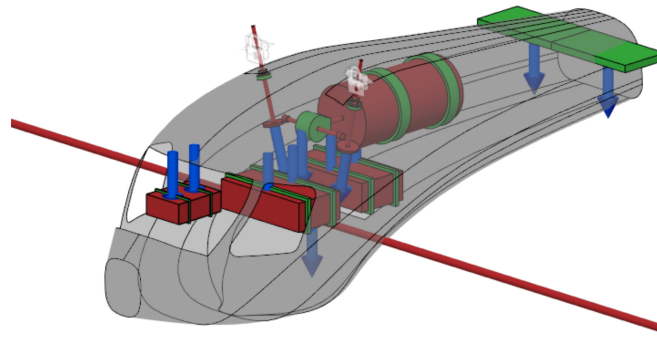


Figure 5: Generative Design Study Setep. Note that an initial fuselage concept shape has been used as a bounding box.

The Generative Design Study and its output has been discussed in further detail in [12]. The result used for initial truss shaping is illustrated in Fig. 6.

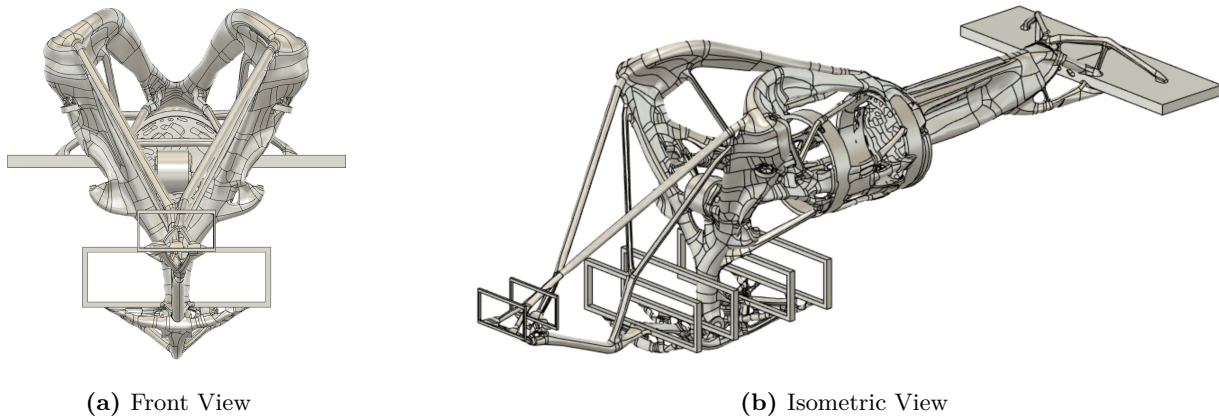


Figure 6: Generative Design Study Output

However, this design could not be used as is, due to its highly organic shape as well as an extremely high mass of approximately 2 tonnes - which was much higher than the initial estimate of 68 kg. However, upon further investigation it was realised that even while using standard Al 7075 Alloy, the structure had a minimum safety factor of 200, leaving a lot of room for manual optimisation.

There were a few key features present in the Generative Design output, that were then used for modelling the actual truss structure:

- Every major component is connected to the base of the rotor by a single member.
- Large members directly connect the base of the rotors to the payload attachment point.
- From the front a triangular shape was obtained. A single beam located at the axis of symmetry of the structure formed the bottom half of the truss.

5.3.3 Initial Truss Modelling

These key features were then used to make a preliminary model of the actual fuselage truss. This model only contained the above-mentioned features and ignored the smaller, more organic features. The whole structure was modelled using several tubes with a constant structure, as these would be easy to procure and assemble. These tubes all had a diameter of 50 mm with a wall thickness of 10 mm, with the exception of the keel beam (The structural member at the bottom of the truss), which was modelled as a solid square cross section with a side length of 40 mm. Fig. 7 shows the preliminary CAD model of the truss structure. Note that the landing gear modelled in this figure is purely to illustrate the landing gear mounting point.

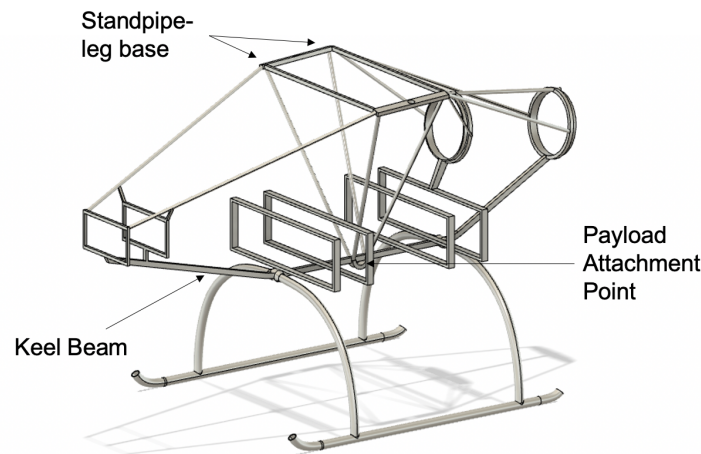


Figure 7: CAD model of the initial truss shape

Note that the tail is not modelled in this section as another group member was assigned to tail boom design. [13] It was decided that the tail boom would fix onto the engine frame, which would transmit tail loads onto the main fuselage truss. To model tail forces in the FE analysis, two moments were applied at the frame - one to account for the weight of the tail structure, and one to account for the lift loads produced by the tail.

As seen in Fig. 7, the fuselage truss is essentially a combination of a pin-jointed frame (the top region) and a beam (the bottom). Furthermore, during this initial design phase, the standpipe-leg structures are not analysed or modelled, as these structures would be under heavy vibrational loading from the rotors, as well as several control forces from the rotors.

5.3.4 Material Selection

The key criteria for material selection were:

- High specific strength (High Yield Stress and Ultimate Tensile Strength).

- High Fatigue Resistance due to vibrational loading imposed by the rotors.
- High fracture toughness to minimise crack growth in the structure.
- High corrosion resistance due to the marine operating environment.

Two aerospace aluminium alloys were shortlisted, Al 7075 and Al 2024. Both are commonly used in aerospace structures, and have high specific strengths, as well as good fatigue resistance. Ultimately however, the Al 7075-T7351 alloy was chosen, as it was a variant of Al 7075 specifically designed to have high corrosion resistance, while retaining its other mechanical properties. The material properties of the alloy are shown in Table 4.

Property	Value
Density	2810 kgm^{-3}
Young's modulus	72 GPa
Yield Stress	435 MPa
Ultimate Tensile Strength	505 Mpa

Table 4: Properties of Al 7075 - T7351 [1]

Although there were several arguments for the use of composites, such as a higher specific strength, fatigue resistance, and corrosion resistance, they were not considered in the material selection process. This was due to the inherent complexity of designing the truss joints if composites were to be used. Methods such as adhesive joints and a combined layup (where the entire truss is laid up together, and then cured, instead of laying up and curing individual members) could be utilized, however these would still

prove to be incredibly challenging to analyze and design under the high stresses and vibrations that the fuselage would experience.

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5.3.5 Manufacturing Method

Traditionally, aerospace Aluminium structures have been riveted and bolted together, as welding Al 7075 and Al 2024 would result in phase segregation at the joint, leading to cracking and ultimately an extremely weak joint. However, riveting and bolting multiple truss members at a single junction would require a heavy, metallic machined joint structure, which would add considerable mass and cost to the structure.

Recent advancements in welding technologies however do make welding of Al 7075 possible, through the infusion of titanium carbide nanoparticles into the filler material, that prevent phase segregation at the joint, with a joint strength of up to 551 MPa. Thus, given the cost and complexity advantages of welding over riveting, it was selected as the manufacturing method for the fuselage truss. [14]

To enhance joint strength and minimise stress concentrations at joints, gusset plates would be used at each joint to provide more surface area for welding the members together. Gusset plates have proven to be very effective at transferring loads in truss structures.

5.3.6 Truss Optimization

The modelled truss had a mass of 150 kg, thus was significantly lighter than the initial shape provided by the generative design study. However, given the truss geometry was modelled with an arbitrary cross section, there was room for further optimization. Given that the structure could not simply be idealised as a pin-jointed structure, due to the presence of a keel beam, frames, and welded joints, the structure was manually optimised using an iterative approach. At each step, the structure was validated using FEA, and suitably modified. A mass optimization target of 80 kg was decided as that was the value predicted by the weights team during the initial MTOW iteration. Furthermore, a safety factor of 1.6 was targeted to follow FAR 29 requirements. This was not enforced at the joints as the model did not account for welding fillets or gusset plates. In the optimization process, joint stresses were simply aimed to be below yield, as in practice, the presence of gussets would greatly reduce stress concentration.

After performing several FEA runs for the structure, it was observed that the keel beam experienced high shear stresses, especially where the landing gear was mounted, and at the locations of the fuel tank and engine frames. Thus, the solid cross section of the keel beam was left solid, and only the circular tubes in the top half of the structure were modified. Finite Element results of the optimized structure are illustrated in Fig. 8.

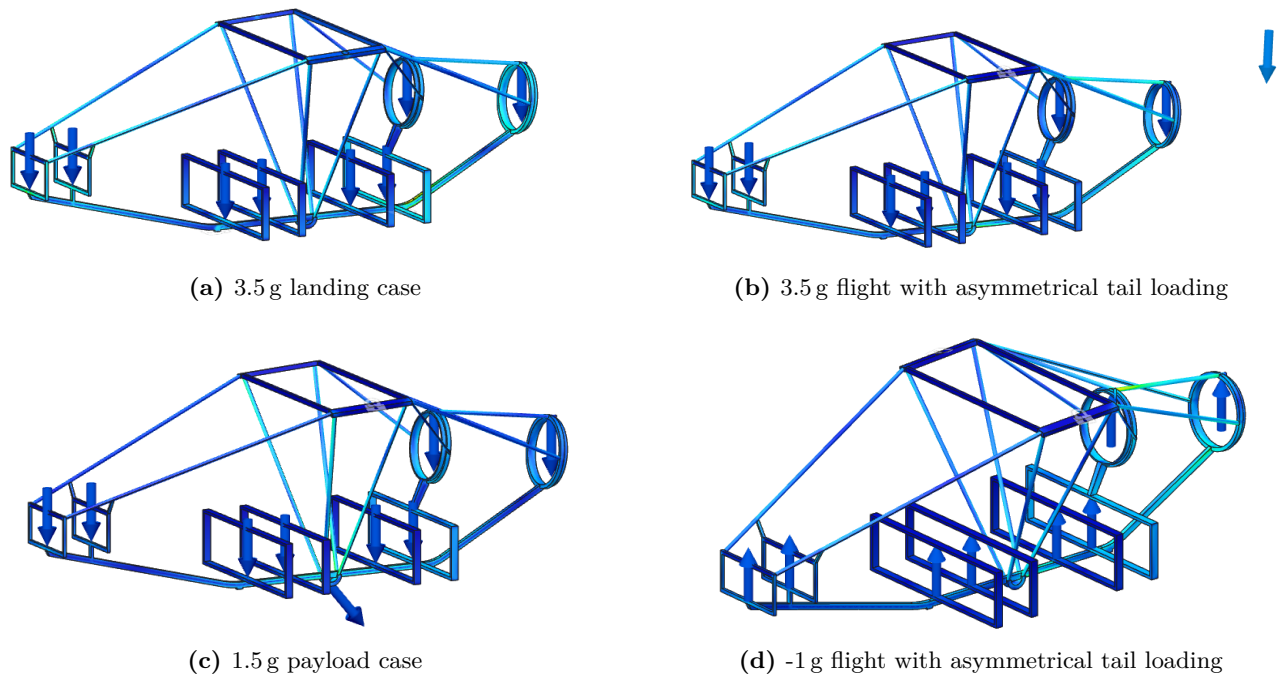


Figure 8: Finite Element Analysis Results for the 4 load cases. The minimum safety factor was 1.3 at the joints, and 1.9 in the members.

The obtained structure was also tested for buckling at the end of the optimization process. The most constraining load case in buckling was the -1 g load case, however its first buckling eigenvalue was 2.191, thus the structure was stable under all load cases.

The optimized structure made use of a solid rectangular cross section keel beam, as well as a circular tubes. Their cross section geometries are shown in Fig. 9

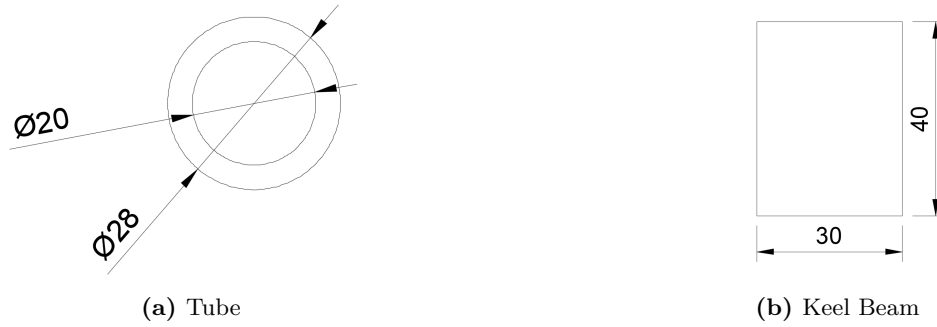


Figure 9: Optimized cross section geometries of tubes and keel beam

5.4 Landing Gear Design

A skid landing gear was selected for the helicopter, due to its mechanical simplicity and low weight. To minimise the weight of the landing gear, it would be designed without any shock absorbing mechanism, and as a single piece to be mounted to the fuselage truss.

5.4.1 Design Constraints and Load Cases

There were several load cases and design constraints stipulated in FAR 29 regulations: [9]

- Have a turnover angle greater than 30° in both longitudinal and lateral directions.
- Fuselage and Landing gear structure to be designed for a load factor produced by a descent velocity of 10 ft/s.
- Skids to be able to withstand a horizontal drag reaction, and a horizontal side load of 50% and 25% of the vertical reaction respectively.
- The skids should be able to handle the above loads in a one skid landing condition. In this case, a rotor lift of $\frac{2}{3} W_{TO}$ can be assumed. [15]

Lastly, a minimum height constraint to the skids was added, such that the rotors could have a minimum height of 1.8 m from the ground plane at the tips. This constraint was added to ensure safety of personnel at the heli-pad.

5.4.2 Material Selection

The primary mechanisms for the dissipation of kinetic energy upon landing are the friction between the skids and the landing pad, and the plastic deformation of the skids. Thus, to ensure high energy absorption, a material with high strain to failure was required. Al 7075-T7351 alloy used for the truss has a strain to failure of 13%, as well as exceptional corrosion resistance, and was thus selected for the landing skids.

5.4.3 Load Factor Estimation

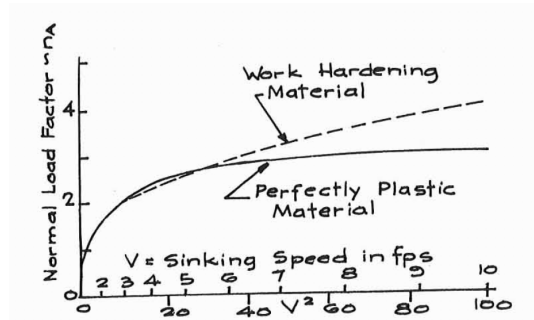


Figure 10: Variation of load factor with descent velocity

The load factor produced upon landing is heavily reliant on landing gear cross tube geometry and surface friction. For this initial design iteration, the load factor was estimated using an empirical method for typical skid landing gears provided in [15]. Fig. 10 highlights the variation in load factor with descent velocity. As the Al 7075 had intermediary properties, the load factor was estimated to be 3.5g for the initial iteration.

5.4.4 Cross Tube Geometry

To maximise specific energy absorption, the cross tubes had a thin walled, circular cross section. Given that analytically calculating optimum dimensions for the cross section proved to be very challenging in the given time frame, initial values were obtained from literature. According to [16], the optimum diameter to thickness ratio for maximum energy absorption was 10. Thus, an initial diameter of 50 mm and a thickness of 5 mm was selected arbitrarily. The cross tubes would have a curved geometry from the frontal view, as this would prevent any stress concentrations in corners during a heavy landing.

5.4.5 Cross Tube Optimization

Given that curved, circular thin-walled beams were used for cross tubes, it was decided that the cross section of the beams would be optimized iteratively, by performing a static and a buckling FEA for the given load cases. Based on these results, the cross sectional diameter and thickness would be suitably modified. After several iterations, during which the cross sectional diameter and thickness of the tubes had to be increased to 60 and 6 mm respectively, an optimized structure is obtained, highlighted in Fig. 11.

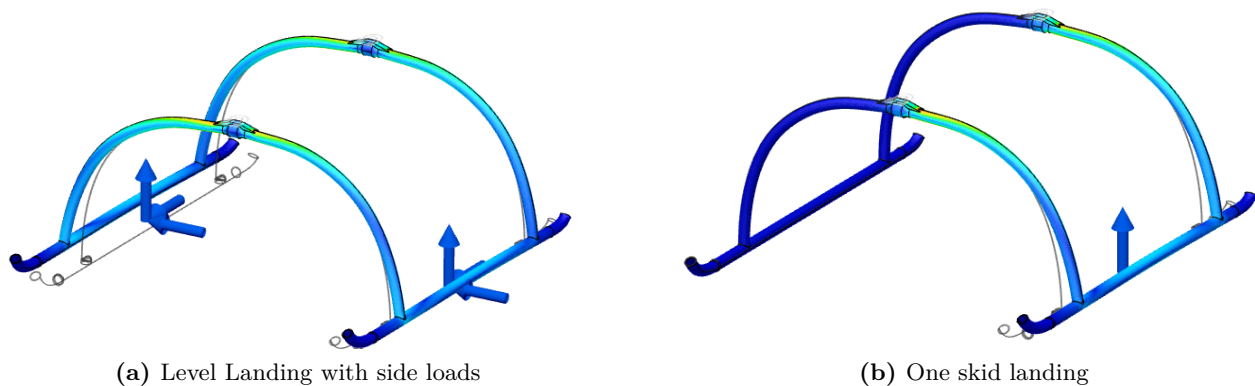


Figure 11: Finite Element Analysis of Skids

The results of the FE simulation suggest that in a level landing case, although the structure does exceed yield stress, it does not exceed the UTS of the material, suggesting that it would deform plastically and

absorb kinetic energy. Exceeding of yield stress is permitted under FAR 29 requirements. However, for the one-skid landing case, the structure does exceed the UTS, as it has a maximum stress of 609 MPa. However, it is to be noted that these are results from a linear static analysis, which does not capture plastic behaviour in regions with high stress concentrations. From Fig. 11b it is clear that only a small zone near the skid mount had an extremely high stress concentration, while the rest of the cross beam was far from failure. Thus, for this design iteration, the high stress concentration was ignored.

5.4.6 Ground Resonance

As the rotor design utilized lead-lag hinges and dampers [17], the helicopter was susceptible to ground resonance. Lead-lag oscillations are usually prevented by lead-lag dampers, however these modes can be triggered by a heavy and unsteady landing. These effects can be minimised by maximising energy absorption by the landing gear. For both the level and one skid landing cases, the cross tubes have been designed to yield to maximise energy absorption. However, to damp out shock loads, dampers would be connected from the keel beam to the cross tubes. These dampers would not take any loads during regular landings, however would serve to damp out oscillations resulting from shock loading of the landing gear.

6 Discussion and Conclusion

Given the limited time frame for this project, the preliminary airframe design only involved one major iteration. Although the airframe was able to achieve a total mass of 95 kg while satisfying FAR 29 structural requirements, there are still major improvements to be made, and several factors that have not been considered. Vibrational loads from the engine, rotors, and transmission were ignored, which if not analysed correctly, could lead to catastrophic failure of the structure. Structural redundancy was also not considered - conventional semi-monocoque airframes have several load paths, such that even if a member fails locally, the entire structure is not compromised.

Reflecting back on the course of the entire project, there were several pitfalls in the design approach taken. Throughout the course of the project, mass and volumes of each component changed several times, leading to delays in finalising the internal layout, which in turn reduced time for structural analysis. Furthermore, the "trial and error" technique used for optimizing the truss and landing gear structure was extremely time consuming, heavily reliant on the CAD model, and might not have resulted in the most optimum solution. As component masses and volumes changed, the entire truss had to be remodelled and retested for the multiple load cases. In hindsight, performing an analytical optimization would have yielded improved results in a much smaller time frame.

Another limitation in the design process was the FE software used. Due to the author not having access to more comprehensive FE packages such as ABAQUS or ANSYS, the FE solver in Fusion 360 was used. Fusion 360 was limited to CST elements, and only linear static stress and buckling analyses could be performed. Using Non-linear static or dynamic analyses, especially for the landing gear, would result in far more accurate results.

Lastly, the aerodynamic skin of the fuselage was not sized, and the contributions of vertical drag onto the airframe loads were ignored. For the next iteration, these would definitely have to be considered.

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